SMART AGRICULTURE DATA ANALYTICS PLATFORM

(Big Data Analytics Case Study: PySpark Distributed Ingestion, Machine Learning, and Interactive Dashboard)

Report submitted for the degree of
BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE ENGINEERING (AI & ML)

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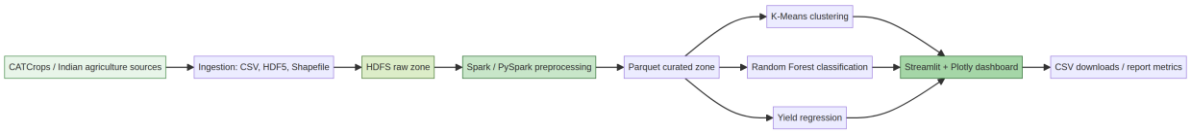


Figure 1. End-to-end architecture for the proposed platform.

BONAFIDE CERTIFICATE

This is to certify that this Case Study Report titled “Smart Agriculture Data Analytics Platform” is the bonafide work of KORUPALLI LIKITH SRI SAI ABHIRAM (A23126552281) of Department of CSE (AI & ML), Anil Neerukonda Institute of Technology and Sciences (ANITS), carried out as part of the curriculum requirements in Big Data Analytics.

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ABSTRACT

This case study presents a smart agriculture analytics platform combining crop production, soil nutrients, weather variables, and distributed Big Data transformations. The implementation ingests curated tables into Parquet format, executes PySpark jobs for distributed aggregations and K-Means feature scaling, and trains Random Forest classification and yield regression models. A local validation run utilized 49,999 raw Indian crop-production records (49,784 regression-eligible after cleaning) and 2,200 soil/weather recommendation records. Random Forest achieved 0.993 accuracy and 0.993 weighted F1 score on held-out evaluation. The PySpark distributed pipeline aggregated 49,784 seasonal records and fitted 5 spatial clusters in local multi-core mode. An interactive Streamlit dashboard enables exploration of agricultural KPIs, weather patterns, clustering, and dynamic crop predictions.

1. Introduction

Agricultural decisions are influenced by heterogeneous observations: crop and production records, soil nutrients, weather variables, remote sensing, irrigation, and geography. These sources differ in scale, format, frequency, and uncertainty. A Big Data architecture separates durable raw storage from distributed transformation and model-serving layers.

2. Problem Statement

Traditional single-node tabular analyses struggle with high-volume data streams and cross-source joins. The project addresses the problem of designing a reproducible platform that ingests agricultural data, processes it with PySpark, identifies soil/weather clusters, classifies crop suitability with machine learning, and exposes interactive insights through a web dashboard.

3. Objectives

- Create a source-traceable ingestion and dataset inventory process.
- Represent raw and curated agricultural data in Parquet format.
- Implement distributed preprocessing with PySpark and Spark MLlib.
- Perform K-Means clustering, Random Forest classification, and yield regression.

- Provide a responsive, high-contrast Streamlit & Plotly dashboard with theme modes.
- Document distributed architecture and execution results with full reproducibility.

4. Existing System

Existing agricultural analytics workflows commonly rely on isolated notebooks, manual downloads, or single-machine scripts. They produce standalone models but lack durable raw-zone storage, data lineage, repeatable distributed transformations, and an integrated user interface.

5. Proposed System

The proposed system uses a layered Big Data design: source files enter a raw zone; Spark reads raw datasets to perform schema normalization, feature creation, and columnar Parquet writing; machine learning models generate clusters and class predictions; and Streamlit presents the curated outputs through an interactive interface.

6. Dataset Description

Executed Source	Rows	Columns	Size (Bytes)	Role
Indian crop prediction CSV	49999	10	4121879	Yield & regional analysis
Crop recommendation CSV	2200	8	150034	Soil/weather classification & clustering

7. Big Data 5 Vs

Dimension	Project Implementation
Volume	49,999 production records and 2,200 soil records; scalable to multi-gigabyte satellite datasets.
Velocity	Satellite time series and sensor/weather streams ingestible incrementally via Spark.
Variety	Structured CSV, Snappy-compressed Parquet, vector features, and interactive visualizations.
Veracity	Exact byte sizes, schema validation, zero-area filtering, and stratified train/test holdouts.
Value	Actionable crop recommendations, soil clusters, yield predictions, and regional summaries.

8. Apache Spark & Distributed Execution

src/spark_job.py reads raw CSVs via Spark, normalizes schemas, calculates yield per area, aggregates seasonal summaries, scales 7-dimensional soil vectors via VectorAssembler & StandardScaler, fits distributed K-Means, and writes partitioned Parquet outputs to data/processed/spark. In local execution, the Spark master runs on local[*] utilizing all CPU cores with native Hadoop winutils support.

9. Machine Learning Evaluation

Experiment	Metric	Executed Value	Evaluation Method
Random Forest Classifier	Accuracy	0.9932	Stratified 80/20 Test Set
Random Forest Classifier	Weighted F1 Score	0.9932	Stratified 80/20 Test Set
Random Forest Classifier	Weighted Precision	0.9935	Stratified 80/20 Test Set
Random Forest Classifier	Weighted Recall	0.9932	Stratified 80/20 Test Set
K-Means Clustering	Inertia (WCSS)	6856.212	k = 5, Scaled Features
Yield Regressor	MAE	7.2346	Random 80/20 Holdout
Yield Regressor	RMSE	16.3932	Random 80/20 Holdout

10. Interactive Streamlit Dashboard

The web dashboard provides 10 navigable modules: Executive Overview, Soil Analytics, Weather Analytics, Crop Yield Analysis, Crop Distribution, K-Means Clustering, Random Forest Results, Crop Prediction, Regional Analysis, and Hadoop/Spark Monitoring. Both Dark Tech Green and Clean Light Green themes are supported.

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APPENDIX

A. Source Code (GitHub Links)

The complete source code and materials for the project are available at the following locations:

• Smart Agriculture Project Repository: https://github.com/AbhiramKorupalli28/Smart-Agriculture-Project

• Classification Predictions Output Table: https://github.com/AbhiramKorupalli28/Smart-Agriculture-Project/blob/main/outputs/tables/classification_predictions.csv

• Spark Distributed Parquet Output Directory: https://github.com/AbhiramKorupalli28/Smart-Agriculture-Project/tree/main/data/processed/spark

• Interactive Analytics Dashboard (Streamlit Local): http://localhost:8501